

Swapnil Meshram

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Summary: Robotics engineer with extensive R&D experience designing embedded systems for autonomous robots, skilled in programming as well as in machine learning and robotics frameworks for perception, control, and planning. Seeking research opportunities to advance collaborative, adaptive, and uncertainty-aware robotic systems for real-world environments.

Education

Master of Science, Robotics and Autonomous Systems (Electrical Engineering)

Arizona State University, Tempe, Arizona

December 2024

GPA: 3.71/4.00

Bachelor of Engineering, Electronics Engineering

K. K. Wagh Institute of Engineering Education and Research

May 2020

Nashik, India

Experience

Arizona State University: Graduate Research Aide

Miniaturized and Advanced Power Electronics Laboratory

February 2023 - September 2023

Tempe, Arizona

- Engineered high-frequency planar transformer PCBs achieving 26 kV and material-enabled 35 kV medium-voltage isolation using 7 kV/mil polyimide for solar thermal applications.
- Co-developed planar transformer PCB designs with Ph.D. researchers for U.S. Department of Energy grant (DE-EE0010239) supporting high-efficiency, low-cost renewable energy integration.
- Configured multi-layer PCB stack-ups (2-14 layers) by optimizing conductor layouts and dielectric spacing to reduce AC copper losses while maintaining robust medium-voltage isolation.
- Resolved critical design challenges by managing fringing electric fields, optimizing via spacing, and refining transformer geometry to minimize conduction losses and ensure successful fabrication.

Aerospace Engineers Private Limited: Founding Electrical Engineer

Autonomous & Undersea Systems Division

February 2021 - December 2022

Tamil Nadu, India

- Spearheaded R&D for maritime autonomous vehicle electrical systems, achieving a 15% cost reduction across key projects including the 300-meter-depth-rated Neerakshi Autonomous Underwater Vehicle (AUV).
- Directed full-cycle embedded system development, managing component selection, schematic capture, PCB layout, and the integration of underwater sensors, mission payloads and wireless communication interfaces.
- Collaborated with cross-functional teams to implement the navigation stack, integrating Artificial Potential Field algorithms and refining Extended Kalman Filters (EKF) for robust state estimation.
- Revamped the AUV's core power distribution system, eliminating critical endurance constraints and reducing internal wiring complexity by 30% to improve system modularity.
- Engineered a high-efficiency propulsion drive, improving thermal management to eliminate active cooling requirements and boost endurance by 5% through optimized Electronic Speed Controller (ESC) tuning.

Projects

Deep Learning Approaches to Musical Instrument Sound Classification

Skills: Python, PyTorch, Librosa, CNNs, Sequence Models, Audio Feature Engineering

January 2024 - April 2024

Tempe, Arizona

- Developed an end-to-end deep learning pipeline for musical instrument classification, achieving 63.75% accuracy with a custom Convolutional Neural Network (CNN) trained on audio spectrograms to extract spatial time-frequency features.
- Benchmarked LSTM, SVM, Transformer, and ResNet-18 models by engineering MFCC and Mel spectrogram pipelines that capture both temporal and spectral dependencies for robust model comparison.

Comparative Analysis of SLAM Algorithms

Skills: ROS, Python, RViz, SLAM, Map Extraction & Error Analysis

January 2024 - April 2024

Tempe, Arizona

- Evaluated Cartographer and GMapping algorithms on a ROS-enabled robot with RPLIDAR, measuring map accuracy via Mean Squared Error (MSE) analysis against actual ground-truth measurements.
- Identified GMapping's superiority in structured environments and Cartographer's strength in complex layouts, recommending depth-camera fusion to resolve 2D LIDAR blind spots caused by low-height obstacles.

EdgeVision: User-Defined Object Counting Using Raspberry Pi

Skills: Python, PyTorch, OpenCV, Deep Learning, Centroid Tracking & Line-Segment Counting

August 2023 - December 2023

Tempe, Arizona

- Engineered a real-time object detection and tracking system on Raspberry Pi 4, optimizing the Faster R-CNN ResNet-50 model to achieve 95.6% accuracy in bright conditions and 82.5% in low-light environments.
- Built custom Centroid Tracking with Euclidean distance matching and an event-driven Line Segment Counter to deliver low-latency, user-defined object counting using virtual counters on dynamic video streams.

Skills

Hardware Design & Simulation: Altium Designer, KiCAD, Autodesk Eagle, Simulink, Ansys Maxwell, LTspice

Programming Languages: Python, C, Embedded C, MATLAB

Development Tools & Environments: ROS1/ROS2, Ubuntu, Google Colab, VS Code, Jupyter Notebook, Anaconda

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn